IBM FileNet P8 V5.6.x

Detailed Study Material

Day 5 – Storage & Security Administration

Storage Configuration | FileNet Security Model | LDAP Integration

Author	Course Day	Version
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Section 1 | Learning Objectives

By the end of Day 5, students will be able to:

1	Describe the FileNet P8 storage architecture including storage areas, storage policies, and content caching
2	Create and configure Fixed Content Device (FCD), file system, and advanced storage areas in ACCE
3	Define and assign storage policies to document classes for automated storage lifecycle management
4	Explain the FileNet P8 security model: Access Control Lists, security inheritance, and role-based access
5	Configure object-level, class-level, and folder-level security using ACCE
6	Integrate WebSphere Virtual Member Manager (VMM) as the LDAP repository for FileNet P8
7	Configure authentication mechanisms: LTPA token-based SSO and form-based authentication via WAS
8	Manage user and group synchronization from LDAP into the FileNet security framework
9	Troubleshoot common storage and security issues using logs and ACCE diagnostics

Section 2 | Storage Configuration

2 Storage Configuration

IBM FileNet P8 uses a layered storage architecture that separates logical storage definitions from physical storage devices. Understanding storage components is essential before deploying any production FileNet system.

2.1 Storage Components Overview

Component	Description
Storage Area	**Logical grouping that maps to a physical storage device (file system, FCD, or object store)**
Storage Policy	Rules that determine which storage area a document is written to based on class or content
Content Cache Area	High-speed temporary storage to accelerate access to frequently retrieved content
Fixed Content Device (FCD)	Write-once-read-many (WORM) compliant storage for regulatory retention
Document Class Storage	Storage area assignment at document class level in ACCE

2.2 Storage Areas

A Storage Area is the fundamental unit of physical storage in FileNet P8. Every document stored in the system is physically located in a storage area. FileNet P8 V5.6.x supports four types of storage areas:

Storage Area Types in FileNet P8 V5.6.x

Type	Description	Use Case
File System Storage Area	Stores document content on a network file system (NFS or CIFS share). Most common type used in training and development environments.	Best For: Development, non-WORM production
Fixed Content Device (FCD)	Integrates with compliant storage hardware/software (IBM Tivoli Storage Manager, EMC Centera, NetApp SnapLock).	Best For: Regulatory compliance, WORM
Advanced Storage Area	Provides policy-driven storage tiering, object store integration, and S3-compatible cloud storage.	Best For: Tiered storage, cloud hybrid
Database Storage Area	Stores small documents (typically <10 MB) directly in the object store database as BLOBs.	Best For: Small files, simplified backup

2.2.1 File System Storage Area – Detailed Configuration

The File System Storage Area is configured in ACCE under the Object Store > Storage Areas node. The following properties must be configured:

Property	Value / Description
Symbolic Name	Unique identifier used internally. No spaces. Example: FS_TrainingStore_01
Display Name	Human-readable name shown in ACCE. Example: Training File Store 01
Root Directory Path	UNC path or local path where content files are stored. Example: C:\FileNetStore\TrainingOS\FS01
Maximum Content Size	Maximum individual file size stored in this area. Default: no limit. Can be set to force large files to alternate areas.
Affinity Cost	Numeric value (0-100). Lower cost = preferred for writes. Used by storage policies to rank areas.
Accessibility	Online (active), Offline (read-only), or Disabled. Only Online areas accept new content.

2.2.2 Fixed Content Device (FCD) Storage Area

FCD storage areas connect FileNet to certified WORM (Write Once Read Many) storage systems. This is required for compliance with regulations such as SEC Rule 17a-4, HIPAA, and SOX.

Key FCD Concepts

- FCD Provider: A Java class that translates FileNet API calls to the specific storage device API.
- FCD Connector: IBM-provided connectors exist for EMC Centera, IBM Tivoli Storage Manager (TSM), NetApp SnapLock, and Hitachi Content Platform.
- Retention Period: Set at the FCD area level. Documents cannot be deleted before the retention period expires.
- ClipID: Unique identifier returned by the FCD device for each stored object. Stored in the FileNet database for retrieval.
- WORM Verification: FileNet verifies compliance by attempting a delete (which is rejected by WORM device) during storage area health checks.

2.3 Storage Policies

Storage policies are rules that determine how and where documents are stored when added to an object store. They enable automatic storage tiering — for example, storing new documents on fast SAN storage then migrating them to cheaper storage after 90 days.

2.3.1 Storage Policy Architecture

A storage policy consists of:

- Initial Storage Area: Where documents are first written
- Destination Storage Area: Where documents are moved after a trigger condition is met
- Trigger Condition: Time-based (age), event-based, or size-based rules
- Migration Schedule: Defines when the policy engine evaluates and migrates content

Policy Property	Description
Policy Name	Unique identifier for the policy. Example: STD_Policy_TrainingOS
Initial Area	Storage area where documents are initially placed on creation
Destination Area	Storage area to which documents are migrated after trigger conditions are met
Trigger Type	Time After Creation, Time After Last Access, or Document Size
Trigger Value	Numeric value for the trigger. Example: 90 (days after creation)
Assignment Scope	Object Store level (default for all classes), Document Class level, or Folder level

2.3.2 Assigning Storage Policies to Document Classes

Storage policies are assigned in ACCE at three levels:

1. Object Store Default: A storage policy assigned at the Object Store level applies to all document classes unless overridden.
2. Document Class Level: Assign a specific policy to override the object store default for a particular class.
3. Instance Level: Applications can specify a storage area per document at creation time via the Content Engine API (IDocument.set_StorageArea()).

2.4 Content Caching

Content caching improves performance by storing frequently accessed documents in a high-speed cache area. When content is requested, FileNet checks the cache first before retrieving from the primary storage area.

2.4.1 Cache Area Configuration

Cache Property	Description
Cache Type	File system-based. The cache area is a local or network file system directory.
Maximum Size	Configurable size limit. When the limit is reached, oldest content is evicted (LRU policy).
Retention Period	How long content stays in cache before automatic removal, even if not at capacity.
Cache Scope	Per-server or shared across a cluster of Application Engine / ICN servers.
Cache Hit Ratio	Monitored via ACCE > Object Store > Storage > Cache Statistics. Target: > 80% for production.

2.4.2 Caching Best Practices

- Place cache areas on local SSD storage for maximum I/O performance
- Size the cache to hold at least 2x the average daily retrieval volume
- Monitor cache hit ratio weekly via ACCE and adjust size based on trends
- Do not place cache areas on WORM storage devices

- In clustered deployments, use separate cache areas per node to avoid contention

Section 3 | FileNet Security Model

3 FileNet Security Model

IBM FileNet P8 implements a comprehensive, object-level security model. Security is enforced at every layer: the database, the Content Engine server, the application layer, and the network layer. Understanding this model is critical for any FileNet administrator.

3.1 Security Architecture

FileNet P8 Security Stack (from bottom to top)

1. Operating System Security: File system permissions on storage area directories
2. Database Security: DB2/Oracle permissions on schema objects (CEADMIN schema)
3. Content Engine Security: Object-level ACLs enforced by the CE server at all API calls
4. Application Layer Security: WAS LTPA tokens and J2EE security roles
5. SSL/TLS Transport: All CE and ICN communication should use HTTPS in production

3.1.1 Security Principal Types

Principal Type	Description
User	An individual identity authenticated via the LDAP repository (VMM in our lab). Example: ceadmin, p8user01
Group	A collection of users defined in LDAP. FileNet resolves group membership dynamically at access time.
Special Identities	#AUTHENTICATED-USERS (any authenticated user), #CREATOR-OWNER (document creator), #EVERYONE (all)

3.2 Access Control Lists (ACLs)

Every securable object in FileNet P8 (documents, folders, object stores, classes) has an Access Control List (ACL). An ACL is a list of Access Control Entries (ACEs), where each ACE grants or denies specific permissions to a security principal.

3.2.1 Access Control Entry Structure

ACE Field	Description
Grantee	Security principal (user or group) this ACE applies to
Access Type	Allow or Deny. Deny ACEs take precedence over Allow ACEs.
Permissions	A bitmask of specific rights (see table below)
Inheritance Flags	Whether this ACE is inherited by child objects

3.2.2 FileNet P8 Permission Levels

Permission Name	Description	API Constant
View Content	Read the binary content (rendition) of a document	FI_VIEW_CONTENT
Read	Read object metadata and properties	FI_READ
Write	Modify object metadata and properties	FI_WRITE
Major Version	Check in a major version of a document	FI_MAJOR_VERSION
Minor Version	Check in a minor version of a document	FI_MINOR_VERSION
Create Child	Create objects inside a container (folder, object store)	FI_CREATE_CHILD
Delete	Delete an object	FI_DELETE
Change Owner	Change the ownership of an object	FI_CHOWN
Change Security	Modify the ACL of an object (most powerful)	FI_CHANGE_STATE
Full Control	All of the above permissions combined	(all bits set)

3.2.3 Viewing and Editing ACLs in ACCE

4. In ACCE, navigate to the Object Store > Browse node
5. Right-click any folder or document > Properties
6. Select the Security tab
7. The current ACL is displayed showing all ACEs
8. Click Add to add a new ACE: search for user/group, select permissions, click OK
9. Click the pencil icon next to an existing ACE to modify permissions
10. Click Remove to delete an ACE from the list
11. Click Save to commit changes

3.3 Security Inheritance

FileNet P8 uses a parent-child inheritance model for security. When an object is created inside a container (folder, object store), it can inherit the ACL from its parent container.

3.3.1 Inheritance Behavior

Concept	Description
Default Template	When object stores or document classes are created, a default ACL template is applied
Inherited ACEs	ACEs marked with the INHERITABLE flag flow down from parent to child automatically
Override	A child object can have its own explicit ACEs in addition to or replacing inherited ones

Propagation	Changes to a parent ACL propagate to all children that have inherited ACEs
Breaking Inheritance	Remove all inherited ACEs from a child to isolate its security from the parent chain

3.3.2 Security Inheritance Levels

- Object Store Level: Top-level security container. All classes and folders inherit from here unless overridden.
- Folder Level: Folders inherit from their parent folder or the root object store. Documents inside a folder inherit from the folder.
- Document Class Level: Document classes have their own ACL that applies to all new documents of that class regardless of folder location.
- Instance Level: Individual documents can have ACEs added directly to override class and folder inheritance.

3.4 Role-Based Access Control (RBAC)

While FileNet uses ACLs at the object level, best practice is to implement Role-Based Access Control by managing groups in LDAP and assigning permissions to those groups rather than individual users.

3.4.1 Recommended Role Structure

Role Name	Permissions	Typical LDAP Group
FileNet_Administrators	Full control of all object stores, storage areas, classes, and security configuration	ceadmin, p8admin
FileNet_ContentManagers	Create, read, update, and delete documents. Cannot change security.	cm_users group
FileNet_Reviewers	Read content and metadata only. View renditions. No create/delete.	review_group
FileNet_RecordsManagers	Manage retention, legal hold, and disposition of records	rm_users group
FileNet_DevelopmentTeam	Create classes, property templates, entry templates. Read all content.	dev_group
FileNet_ReportingUsers	Read-only metadata access for reporting and analytics	report_group

Section 4 | LDAP Integration with WebSphere VMM

4 LDAP Integration with WebSphere Virtual Member Manager

IBM FileNet P8 does not store user credentials itself. Instead, it delegates all authentication to the J2EE application server — WebSphere Application Server (WAS) in our training environment. WAS uses a federated user repository called Virtual Member Manager (VMM) to connect to one or more LDAP directories.

4.1 Authentication Architecture

The following diagram describes the authentication flow in FileNet P8 V5.6.x with WAS VMM:

Authentication Flow: User Login to FileNet

Step 1: User opens ICN or ACCE in browser - HTTP request arrives at WAS
Step 2: WAS checks for a valid LTPA token cookie (SSO). If valid, session continues.
Step 3: If no LTPA token, WAS redirects to login page (form-based or BASIC auth)
Step 4: User submits credentials. WAS VMM binds to LDAP to verify username/password.
Step 5: On success, WAS issues an LTPA token (encrypted, time-limited cookie)
Step 6: The LTPA token is passed with every subsequent API call to Content Engine
Step 7: CE receives the LTPA token, extracts the authenticated username
Step 8: CE looks up the user group memberships from VMM attribute store
Step 9: CE evaluates ACLs against the user + group memberships - access is granted or denied

4.2 WebSphere VMM Architecture

WebSphere VMM (Virtual Member Manager) is a WAS component that provides a unified view of users and groups from one or more identity repositories. In our training environment, VMM is configured with one LDAP repository.

4.2.1 VMM Key Concepts

VMM Concept	Description
Federated Repository	VMM's top-level container that federates one or more identity repositories into a single namespace
Realm	The security domain name. All users in VMM belong to a realm. Example: defaultWIMFileBasedRealm
LDAP Repository ID	Unique identifier within VMM for the LDAP connection. Example: TrainingLDAP
Base DN	The Distinguished Name at which VMM starts searching for users and groups in the LDAP tree
Bind DN	The service account VMM uses to connect to LDAP for search operations
Group Member Attribute	LDAP attribute that stores group members. For OpenLDAP: member. For AD: member or memberOf.
User Filter	LDAP filter used to search for user objects. Example: (&(uid={0})(objectClass=inetOrgPerson))

4.3 Configuring LDAP in WebSphere VMM

All VMM configuration is performed through the WAS Admin Console. The following is the complete step-by-step procedure used in our training lab.

4.3.1 Pre-Configuration Checklist

Before Starting VMM LDAP Configuration

Verify: LDAP server is running and accessible from the WAS host on port 389 (LDAP) or 636 (LDAPS).
Verify: Bind DN service account exists and has read access to the user and group OUs.
Verify: The Base DN, user search filter, and group member attribute are confirmed with your LDAP admin.
In training lab: The LDAP server embedded in WAS is used. Host: localhost, Port: 389.
Training Bind DN: uid=wpsbind,o=defaultWIMFileBasedRealm
Training Base DN: o=defaultWIMFileBasedRealm

4.3.2 Step-by-Step: Add LDAP Repository to VMM

12. Open WAS Admin Console: https://localhost:9043/ibm/console
13. Log in as wasadmin
14. Navigate to: Security > Global Security
15. In the 'User account repository' section, click Federated repositories (if not already selected, select it and click Set as current)
16. Under 'Related Items', click Manage Repositories
17. Click Add > LDAP repository
18. Set Repository Identifier: TrainingLDAP
19. Set Primary host: localhost, Port: 389
20. Set Base entry distinguished name: o=defaultWIMFileBasedRealm
21. Set Bind DN: uid=wpsbind,o=defaultWIMFileBasedRealm
22. Set Bind password: (configured during WAS VMM setup)
23. Set LDAP entity type: InetOrgPerson (for OpenLDAP) or Person (for Active Directory)
24. Click OK, then Save to master configuration

4.3.3 Configure Federated Repository to Include LDAP

25. Navigate back to: Security > Global Security > Federated repositories
26. Click the federated repository name (defaultWIMFileBasedRealm)
27. Under 'Repositories in the realm', click Add Base entry to Existing Repository
28. Select the TrainingLDAP repository from the dropdown
29. Set Base entry distinguished name in the repository: o=defaultWIMFileBasedRealm
30. Set Base entry distinguished name in the federated repository: o=defaultWIMFileBasedRealm
31. Click OK and Save
32. Click Test Connection next to TrainingLDAP to verify connectivity

4.4 Authentication Mechanisms

4.4.1 LTPA Token-Based SSO

LTPA (Lightweight Third-Party Authentication) is the default WAS authentication mechanism. It enables Single Sign-On (SSO) across multiple WAS applications on the same domain.

LTPA Concept	Description
LTPA Token Format	Encrypted, base64-encoded cookie containing username, expiry, and digital signature
LTPA Key	Symmetric encryption key used to sign and verify tokens. Must be the same on all WAS nodes in a cluster.
Token Expiry	Default: 30 minutes. Configurable in WAS Admin Console under Security > Global Security > LTPA.
SSO Domain	The cookie domain for which the LTPA token is valid. Example: .ibm.com enables SSO for all *.ibm.com apps.
Token Validation	Content Engine validates the LTPA token on every API call by decrypting it with the shared LTPA key.
LTPA Key Export	Export the LTPA key from WAS for distribution to other WAS nodes: Security > Global Security > LTPA > Export keys.

4.4.2 Form-Based Authentication

Form-based authentication presents an HTML login form to the user instead of a browser-native BASIC auth dialog. This is the standard for ICN and ACCE in production.

- Configured in WAS via: Applications > WebSphere Enterprise Applications > [AppName] > Security role to user/group mappings
- The login-config element in web.xml specifies FORM as the auth-method and the login page URL
- After successful login, WAS issues an LTPA token and redirects the user to the originally requested URL

4.5 Group Management

Groups in FileNet P8 are always defined in the LDAP directory (via VMM). The FileNet Content Engine does not store group definitions — it queries VMM at runtime for group membership.

4.5.1 Group Lookup Flow

33. A user submits an API request to Content Engine
34. CE extracts the authenticated username from the LTPA token
35. CE calls VMM's group membership API: getMembership(username)
36. VMM queries the LDAP repository for all groups the user belongs to (including nested groups)
37. VMM returns a list of group distinguished names
38. CE evaluates the request against the object's ACL using the username + group list

4.5.2 Nested Group Support

FileNet P8 supports nested group resolution when the LDAP server supports recursive group lookup.
Configuration:

- In WAS VMM LDAP configuration, set 'Recursive search for group membership' to true
- Group nesting depth is configurable via the ibm-allGroups attribute in WAS VMM
- Active Directory supports nested groups natively via the memberOf:1.2.840.113556.1.4.1941: filter
- OpenLDAP requires memberOf overlay to be configured for recursive group lookup

4.5.3 Recommended LDAP Group Structure for FileNet

LDAP Group DN	Purpose & Members
cn=filenet_admins,ou=groups,o=defaultWIMFileBasedRealm	Maps to FileNet_Administrators role. Contains ceadmin, p8admin.
cn=filenet_cm_users,ou=groups,o=defaultWIMFileBasedRealm	Content managers with create/read/update/delete on documents.
cn=filenet_reviewers,ou=groups,o=defaultWIMFileBasedRealm	Read-only access to browse and retrieve content.
cn=filenet_rm_users,ou=groups,o=defaultWIMFileBasedRealm	Records managers with access to manage retention and holds.

Section 5 | Troubleshooting Reference

5 Troubleshooting Storage & Security

5.1 Storage Troubleshooting

Symptom	Likely Cause	Resolution
Content not found on retrieval	Storage area offline or root directory path changed after content was stored	Set storage area back Online in ACCE. Verify root directory path is accessible.
Disk full on storage area	Storage area root directory has no remaining disk space	Add a new storage area and update the storage policy to redirect new content. Migrate existing content offline.
Storage policy not applied	Policy was not assigned at the correct level (class vs. object store)	Verify assignment in ACCE > Object Store > Storage Policies. Check document class properties.
Cache area not caching	Cache area is disabled or max size is set too low	Set cache area to Online. Increase max size. Check %FILENET_IS_HOME%\logs\IS_server_log for cache errors.
FCD content not retained	FCD provider configuration is incorrect or retention period is set to 0	Verify FCD provider class, connection parameters, and retention period in ACCE.

5.2 Security Troubleshooting

Symptom	Cause	Resolution
User cannot access document	User or their groups not in the document ACL	Check ACCE > document > Security tab. Add the user's group with the required permissions.
LDAP authentication fails	VMM LDAP connection error or bind DN credentials expired	Test connection in WAS Admin Console > Manage Repositories. Verify bind DN password.
User not found in FileNet	User exists in LDAP but not visible in CE/ICN user search	Verify the user search filter in VMM LDAP config includes the user's objectClass.
Groups not resolved	Nested group resolution disabled or LDAP server does not support recursive lookup	Enable recursive group search in VMM LDAP config. Test with ldapsearch from the WAS host.
LTPA token invalid	LTPA keys out of sync between WAS nodes or token has expired	Re-export and re-import LTPA keys across all WAS nodes. Verify token timeout setting.

Access denied despite ACL allow	A Deny ACE on a parent object or group is overriding the Allow ACE	Check all parent objects in the path for Deny ACEs. Deny takes precedence over Allow.

5.3 Key Log Files

Log Location	Contents
%FILENET_IS_HOME%\logs\IS_server_log	Content Engine server log. Contains CE startup, API errors, and storage access errors.
%WAS_HOME%\profiles\FileNetServer\logs\server1\SystemOut.log	WAS server log. Contains authentication failures, VMM errors, and LTPA token issues.
%WAS_HOME%\profiles\FileNetServer\logs\server1\SystemErr.log	WAS error stream. Contains Java exceptions and stack traces.
ACCE > Object Store > Health Monitoring	ACCE built-in health monitor. Shows storage area status, replication errors, and event errors.
WAS Admin Console > Troubleshooting > Logs and Trace	Enable LDAP trace: com.ibm.ws.wim.* = finest for VMM debugging.

5.4 Quick Reference: Key Values for Training Lab

Item	Value	Notes
WAS Admin Console	https://localhost:9043/ibm/console	wasadmin / wasadmin
ACCE URL	http://localhost:9080/acce	ceadmin / ceadmin
Object Store	TrainingOS	Created on Day 2
CE Web Services	http://localhost:9080/wsi/FNCEWS40MTOM/	MTOM endpoint for API
LDAP Host	localhost	Embedded WAS LDAP (VMM)
LDAP Port	389 (clear) / 636 (SSL)	Use 389 in lab
VMM Base DN	o=defaultWIMFileBasedRealm	Default WAS VMM realm
Training Storage Path	C:\FileNetStore\TrainingOS\	Create before lab
Log Path (CE)	C:\IBM\FileNet\ContentEngine\logs\	IS_server_log